
Quantile-Initialised Ordinal Collaborative Filtering with Gated Auxiliary Fusion

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Abstract

We propose a simple calibration recipe for cumulative-link collaborative filtering: initialise the monotone thresholds from the empirical rating quantiles of the training set, parameterise their spacing through a softplus-inverse, and exclude them from L_2 regularisation. The recipe matches the data marginal at training start and gives a native rating distribution without sacrificing RMSE. Paired with per-dimension gated fusion of side information, the gated+ordinal configuration attains RMSE 0.9124 ± 0.0036 and NLL 1.2584 across the five canonical MovieLens-100K splits, beating both a no-fusion MF analogue (0.9160) and additive fusion (0.9189) on the across-split RMSE mean. A MovieLens-1M scale-up preserves the same ordering (RMSE 0.8481 ± 0.0006 , NLL 1.184). The contribution is practical rather than algorithmic: standard components are combined into a reproducible, calibrated recommender pipeline.

1 Introduction

Collaborative filtering (CF) for explicit feedback predicts unobserved user-item ratings from a sparse matrix. Latent-factor models based on matrix factorisation [4] remain standard baselines and building blocks on MovieLens [2]. Two persistent issues remain when these models are applied as a course-style data-analytics tool rather than a production recommender. First, side information (user demographics, item genres) is typically injected by adding a projected feature vector to the latent embedding, with no learned mechanism to decide *when* to trust the embedding versus the side information. Second, the rating is treated as a continuous quantity even though it is in fact a five-class ordinal variable; the resulting predictions can be RMSE-competitive yet badly calibrated as distributions over star ratings.

Related work and contribution. The two issues above have been addressed separately: gated content fusion is used by GATE [6], while cumulative-link ordinal regression starts from McCullagh’s model [7] and appears in recommenders such as OrdRec [5], OrdNMF [1], and OPRFM [9]. Our contribution is implementation-level rather than algorithmic: we combine gated side-information fusion with a calibrated cumulative-link head on a transparent MF core, and we separate their effects through ablations, calibration plots, cold-start analysis, and a small MovieLens-1M scale-up.

2 Problem Definition

Let $\mathcal{U} = \{1, \dots, n\}$ be the set of users and $\mathcal{I} = \{1, \dots, m\}$ the set of items. Observed ratings form a sparse set $\mathcal{D} = \{(u_t, i_t, r_t)\}_{t=1}^N$ with $r_t \in \{1, 2, 3, 4, 5\}$. Each user u carries a side-information vector $f_u \in \mathbb{R}^{d_u}$ (z-scored age, one-hot gender, one-hot occupation in MovieLens-100K, $d_u = 24$) and each item i carries $g_i \in \mathbb{R}^{d_i}$ (multi-hot genre indicator, $d_i = 19$). The task is to predict $\hat{r}_{ui}$ for

held-out (u, i) pairs, evaluated by RMSE, MAE, classification accuracy of the rounded prediction, and where applicable the held-out negative log-likelihood (NLL) of the predicted rating distribution.

3 Method

3.1 Matrix factorisation backbone

We start from a biased dot-product MF model. Each user and item carries a d -dimensional embedding $(p_u, q_i \in \mathbb{R}^d)$ and a scalar bias $(b_u, b_i \in \mathbb{R})$. The pre-head score is

$$s_{ui} = p'_u \cdot q'_i + b_u + b_i, \quad (1)$$

where p'_u, q'_i are the (possibly fused) representations defined below. With fusion disabled, $p'_u \equiv p_u$ and $q'_i \equiv q_i$.

3.2 Gated auxiliary fusion

Let $a_u = \text{ReLU}(W_u f_u + b_{f,u})$ be a learned projection of the user side features into the embedding space, and similarly $a_i = \text{ReLU}(W_i g_i + b_{f,i})$ for items. We define the fused user representation as

$$g_u = \sigma(W_g [p_u \parallel a_u] + b_g), \quad p'_u = (\mathbf{1} - g_u) \odot p_u + g_u \odot a_u, \quad (2)$$

where $\parallel$ denotes concatenation and $\odot$ is the Hadamard product, so the gate $g_u \in [0, 1]^d$ acts *per dimension*. The item-side fusion is identical with (q_i, a_i) in place of (p_u, a_u) . We zero-initialise both W_g and b_g , which makes $g_u \equiv \frac{1}{2}\mathbf{1}$ at training start. This gives every dimension the same neutral starting mixture between the ID embedding and the ReLU-projected side features; the gate then opens and closes per dimension as gradient flows. As an ablation we also consider the *additive* variant $p'_u = p_u + a_u$ and the *none* variant $p'_u \equiv p_u$ (and similarly for items).

3.3 Ordinal regression head

For the head we treat r as an ordinal variable with $K = 5$ classes. We use the cumulative-link model of McCullagh [7] with monotone thresholds $\theta_1 < \theta_2 < \theta_3 < \theta_4$:

$$P(r \leq k \mid s) = \sigma(\theta_k - s), \quad P(r = k \mid s) = \sigma(\theta_k - s) - \sigma(\theta_{k-1} - s), \quad (3)$$

with $\theta_0 = -\infty$ and $\theta_5 = +\infty$. We enforce monotonicity by the softplus reparameterisation

$$\theta_k = \theta_1 + \sum_{j=1}^{k-1} \text{softplus}(\delta_j), \quad (4)$$

which guarantees $\theta_k < \theta_{k+1}$ for any unconstrained parameters.

Empirical-quantile initialisation (the recipe). We initialise from the empirical rating distribution. Let $\hat{p}_k$ be the training-set frequency of rating k and $\hat{F}_k = \sum_{j \leq k} \hat{p}_j$ its cumulative. Choosing

$$\theta_k = \text{logit}(\hat{F}_k), \quad \delta_j = \log(\exp(\theta_{j+1} - \theta_j) - 1) \quad (5)$$

guarantees $P(r \leq k \mid s=0) = \hat{F}_k$, so the predicted marginal at the start of training equals the data marginal exactly. The second equation is just the softplus inverse, recovering the unconstrained δ_j that the optimiser then updates. We also exclude $\{\theta_1, \delta_j\}$ from weight decay because their scale carries calibration information.

The training loss is the negative log-likelihood $\mathcal{L}_{\text{ord}} = -\frac{1}{N} \sum_t \log P(r = r_t \mid s_{u_t i_t})$, and the point prediction is the expected rating $\hat{r}_{ui} = \sum_{k=1}^5 k P(r = k \mid s_{ui})$. As an ablation we also consider the standard *sigmoid* head: $\hat{r}_{ui} = 4\sigma(s_{ui}) + 1$ trained with mean-squared error.

3.4 Complexity

The MF backbone has $(n + m)d$ embedding parameters plus $n + m$ biases; for MovieLens-100K with $n = 943$, $m = 1682$, $d = 128$, that is $\approx 3.4 \times 10^5$ parameters. Each fusion module adds a projection $(d_{u/i} \cdot d + d)$ and a gate $(2d^2 + d)$; summed over both sides this is $\approx 7 \times 10^4$ parameters — about one-fifth of the embedding budget. The ordinal head adds only $K - 1 = 4$ parameters, and the dominant per-rating cost remains $O(d)$.

4 Experiments

4.1 Setup

We use MovieLens-100K [2] on its five canonical 80/20 splits `u1–u5`, each yielding 80,000 training and 20,000 test ratings on the same 943 users and 1,682 items. Headline numbers are reported as mean $\pm$ std across the five per-split means (each split-mean itself averages three training seeds). The visualisation and cold-start analyses are restricted to the `u1` split. User side features are z-scored age, two-class one-hot gender, and 21-class one-hot occupation ($d_u = 24$); item features are 19-genre multi-hot ($d_i = 19$). All experiments use Adam with learning rate 10^{-3} and weight decay $\lambda = 10^{-5}$ on all parameters except the ordinal thresholds. Early stopping uses validation RMSE only; the test set is read once after restoring the best-validation weights.

4.2 Baselines and metrics

We compare against truncated SVD, biased MF trained with MSE, and NMF. Metrics are RMSE, MAE, rounded-rating accuracy, and test NLL where a native class-probability head is available.

4.3 Results

Table 1 places the proposed model against transparent baselines on the `u1` split. The stronger evidence is the multi-split ablation in Table 2.

Table 1: Transparent baselines on MovieLens-100K `u1`. Lower is better for RMSE/MAE/NLL, higher for Accuracy.

Model	RMSE	MAE	Accuracy	NLL
SVD (rank 20)	1.0834	0.8896	0.3716	—
Biased MF	0.9221	0.7274	0.4124	—
NMF	0.9197	0.7252	0.4146	—
Gated+ordinal	0.9122	0.7130	0.4367	1.2472

The ablation matrix in Table 2 reports the full $\{\text{none, additive, gated}\} \times \{\text{sigmoid, ordinal}\}$ cross-product. The proposed configuration has the best mean value on all four reported metrics. It improves RMSE by 0.0036 over the no-fusion sigmoid baseline and by 0.0065 over additive fusion with a sigmoid head. Replacing only the head from sigmoid to ordinal on the same gated backbone gives a small RMSE change (+0.0011) but improves MAE by 0.0056, accuracy by 0.0087, and provides a native class distribution (NLL 1.2584). We therefore read the ordinal head mainly as a calibration and per-class-behaviour mechanism rather than as the main source of RMSE gain.

Table 2: Ablation matrix on MovieLens-100K, mean $\pm$ std across the five canonical splits `u1–u5` (each split-mean averages three seeds). Bold = best per column. Lower is better for RMSE/MAE/NLL, higher for Accuracy.

Fusion	Head	RMSE	MAE	Accuracy	NLL
none	sigmoid	0.9160 $\pm$ 0.0061	0.7241 $\pm$ 0.0059	0.4233 $\pm$ 0.0044	—
none	ordinal	0.9170 $\pm$ 0.0072	0.7204 $\pm$ 0.0069	0.4295 $\pm$ 0.0056	1.2634 $\pm$ 0.0050
additive	sigmoid	0.9189 $\pm$ 0.0044	0.7238 $\pm$ 0.0055	0.4243 $\pm$ 0.0057	—
additive	ordinal	0.9174 $\pm$ 0.0040	0.7173 $\pm$ 0.0035	0.4340 $\pm$ 0.0038	1.2654 $\pm$ 0.0017
gated	sigmoid	0.9135 $\pm$ 0.0037	0.7182 $\pm$ 0.0034	0.4286 $\pm$ 0.0054	—
gated	ordinal	0.9124 $\pm$ 0.0036	0.7126 $\pm$ 0.0042	0.4373 $\pm$ 0.0045	1.2584 $\pm$ 0.0043

4.4 Ablation study

Table 2 supports three claims.

Gated fusion gives the larger RMSE change. Holding the head fixed, replacing *none* fusion with *gated* drops RMSE by 0.0025 (sigmoid) or 0.0046 (ordinal); replacing *additive* with *gated* drops it by 0.0054 (sigmoid) or 0.0050 (ordinal). Additive fusion is consistently the worst sigmoid-head cell, suggesting that unconditional summation of side information can hurt once bias terms absorb marginal effects.

The ordinal head buys native probabilities; its RMSE effect depends on fusion. Switching from sigmoid to ordinal moves RMSE by at most 0.0015 in any fusion regime, all within between-split std. What it reliably buys is native rating probabilities and per-class accuracy: it improves MAE by 0.003–0.007 and accuracy by 0.004–0.010 in every fusion regime.

Ordinal NLL is low across all ordinal cells. The proposed configuration achieves NLL 1.2584, well below the log 5 ≈ 1.609 uniform baseline. The quantile initialisation is the mechanism: the predicted marginal matches the data marginal at $s = 0$ to numerical precision.

Reliability diagram and per-class ECE. Figure 1 renders the calibration curve directly. The ordinal head’s ECE is lower on four of the five classes (1, 2, 3, and 5), with the kernelised sigmoid winning only on the dominant class 4. This matches the per-class F1 pattern: sigmoid+MSE concentrates mass around the centre of the rating range, while the ordinal head spreads probability across the five classes more evenly.

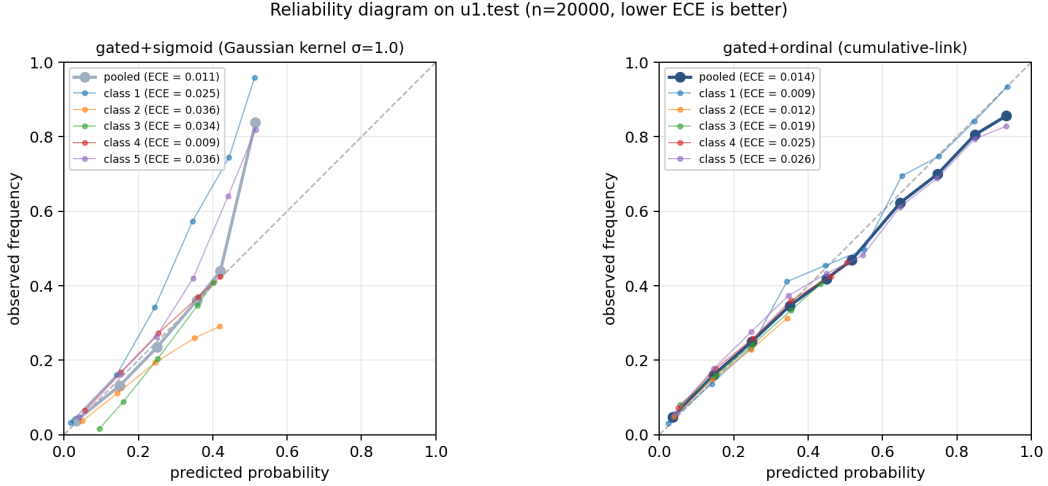


Figure 1: Reliability diagrams on `u1.test`. Gated+ordinal is better calibrated on classes 1, 2, 3, and 5; kernelised gated+sigmoid wins only on the dominant class 4.

Where the ordinal head wins: extreme classes. Figure 2 reports per-class F1 on `u1.test`. The ordinal head wins on the boundary classes: +0.098 F1 on class 1 and +0.035 F1 on class 5; classes 2–4 are within ± 0.015 . Macro-F1 lifts from 0.341 to 0.369, driven mainly by recall on classes 1 and 5. This is the per-class footprint of the calibration claim.

Sensitivity checks supported $d = 128$ and $\lambda = 10^{-5}$: increasing to $d = 256$ improved RMSE by only 0.0024 at +34% wall time, while $\lambda = 10^{-4}$ overregularised by 0.0080 RMSE.

4.5 Embedding visualisation

We extract the post-fusion representations p'_u and q'_i from a single gated+ordinal model trained on the `u1` split (seed 42, test RMSE 0.9165), and project them to $\mathbb{R}^2$ via three classical manifold methods: PCA [3] (linear, variance preserving), classical MDS (linear, distance preserving), and IsoMap [8] (nonlinear, geodesic distance preserving, $k = 15$ neighbours). To keep figures legible, we sample ≈ 200 users balanced across the six most common occupations (+ “rest”), and ≈ 200

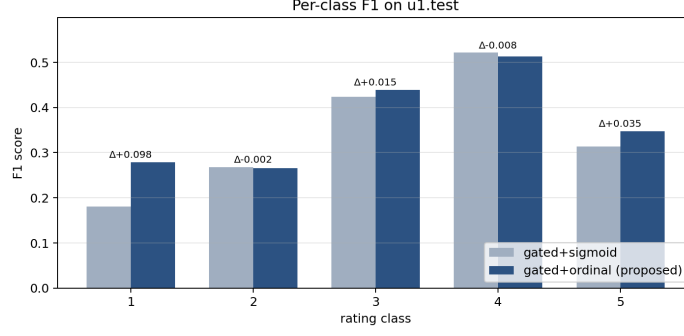


Figure 2: Per-class F1 on `u1.test` for the same gated backbone with sigmoid vs. ordinal heads. The ordinal head wins on the extreme classes (1 and 5) where the cumulative-link thresholds carry boundary information; classes 2–4 are within seed-level noise.

items balanced across the six most common dominant genres (+“rest”). Cluster quality is measured by the silhouette score with respect to the categorical labels.

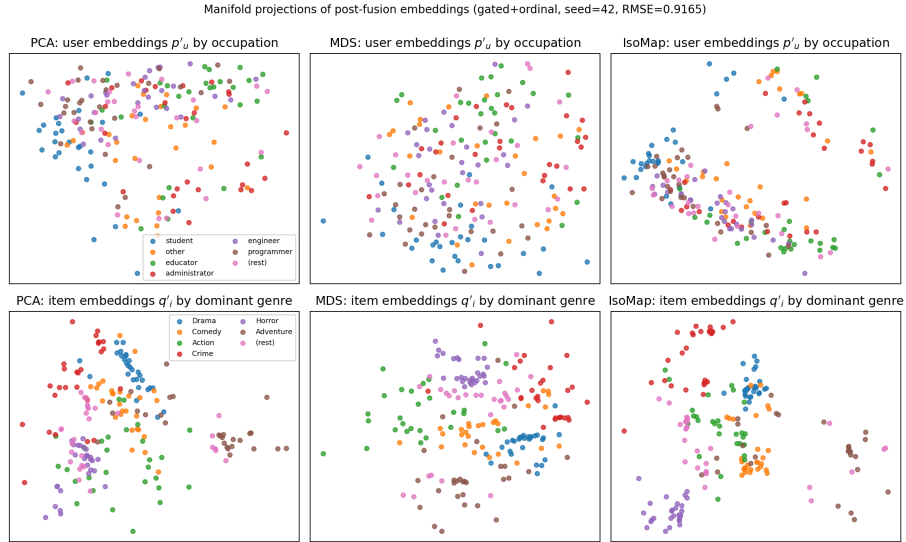


Figure 3: Two-dimensional projections of the post-fusion user representations p'_u (top, coloured by occupation) and item representations q'_i (bottom, coloured by dominant genre). Columns show PCA, classical MDS, and IsoMap with $k = 15$ neighbours. Item-side genre structure is recovered most clearly by IsoMap; user-side occupation structure is present in the high-dimensional space but does not survive projection to two dimensions.

Table 3: Silhouette scores in the full 128-dimensional space (HD) and after projection to two dimensions, with respect to occupation labels (users) and dominant genre labels (items).

	HD ($d = 128$)	2D PCA	2D MDS	2D IsoMap
p'_u by occupation	+0.0848	−0.107	−0.103	−0.114
q'_i by dominant genre	+0.1930	+0.048	+0.137	+0.232

The model learns side-information structure without an auxiliary classification loss. Item-side genre structure is clearer than user-side occupation structure in 128-D (+0.193 vs +0.085), and it survives nonlinear projection better than linear projection (IsoMap +0.232 vs PCA +0.048). Occupation information is present in high dimension but does not compress cleanly into two dimensions.

4.6 Gate behaviour and cold-start regime

Per-user trajectory. We stratify test predictions by the corresponding user’s training activity $|R_u|$ into five buckets and compare gated+ordinal against additive+ordinal.

Table 4: Cold-start RMSE breakdown by user activity. Negative Δ means additive fusion wins on that bucket.

$ R_u $ bucket	< 30	30–59	60–119	120–239	≥ 240
gated RMSE	0.9799	0.9649	0.8841	0.8896	0.9200
additive RMSE	0.9803	0.9768	0.8929	0.8946	0.9128
Δ (add. – gated)	+0.0004	+0.0119	+0.0088	+0.0051	–0.0071
mean g_u	0.246	0.256	0.263	0.268	0.291
mean g_i	0.316	0.317	0.316	0.317	0.316

The cold-start table complicates the aggregate result. Gated fusion is strongest for mid-activity users, ties on the coldest bucket, and reverses for the 14 most active users. Its gain is therefore not uniform across activity regimes, even though it wins on the overall mean.

4.7 Gate interpretability

Aggregate gate values disguise a useful per-dimension pattern. For the trained gated+ordinal model on `u1`, Figure 4 sorts each dimension by its population-mean gate value. The user-side gate spans $[0.002, 0.984]$ with mean 0.263; the item-side gate spans $[0.002, 0.992]$ with mean 0.317. Only about one fifth of dimensions sit above the zero-init value of 0.5, so the fusion mechanism is not simply averaging ID embeddings and side features. Most dimensions specialise toward the ID pathway, while a smaller subset opens strongly toward side information.

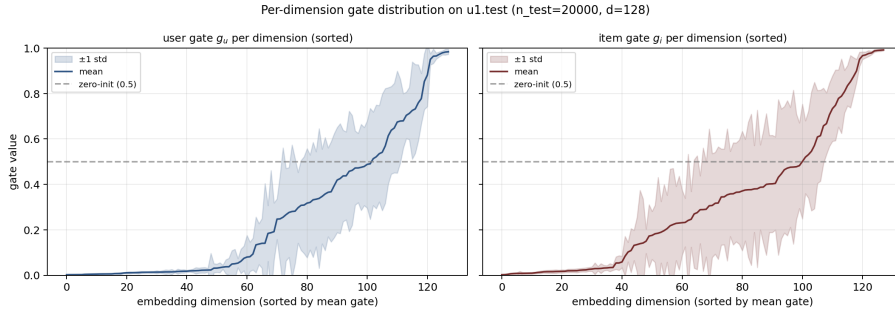


Figure 4: Per-dimension gate distribution on `u1.test`, sorted by mean. The shaded band is ± 1 std across the test population.

4.8 Scale-up to MovieLens-1M

To check scale, we run the three most informative cells on MovieLens-1M (1,000,209 ratings) under a deterministic random 80/10/10 split. All other settings match the MovieLens-100K experiments, and each cell averages the same three training seeds.

Table 5: Scale-up to MovieLens-1M, random 80/10/10 split, mean $\pm$ std over three seeds.

Fusion	Head	RMSE	MAE	Accuracy	NLL
none	sigmoid	0.8567 ± 0.0008	0.6782 ± 0.0007	0.4480 ± 0.0005	—
gated	sigmoid	0.8502 ± 0.0003	0.6695 ± 0.0001	0.4579 ± 0.0002	—
gated	ordinal	0.8481 ± 0.0006	0.6630 ± 0.0010	0.4654 ± 0.0014	1.1842 ± 0.0013

The ordering matches MovieLens-100K: gated fusion improves RMSE over no fusion, and adding the ordinal head improves it again. The larger data set also tightens seed variance, consistent with more stable gate training.

4.9 Reproducibility

The code path is intentionally small enough to audit. All reported tables and figures are regenerated from archived JSON/CSV outputs or from the scripts in Table 6. Each script records the model configuration, split, seed, and metric outputs used in the paper. The reported ablation uses training seeds {42, 43, 44} on each canonical split and fixes the validation-split seed so split variance is not mixed with model-initialisation variance.

Table 6: Core reproduction scripts. Wall times are approximate on M1 Pro CPU.

Result	Script	Time
Main ablation, 6 cells, 5 splits	<code>run_multisplit.py</code>	~60 min
Baseline sanity check	<code>run_baselines.py</code>	~5 min
Sensitivity curves	<code>run_sensitivity.py</code>	~15 min
Class F1 and reliability	<code>run_classmetrics.py, run_reliability.py</code>	~4 min
Embedding and cold-start diagnostics	<code>run_visualization.py, run_coldstart.py</code>	~3 min
MovieLens-1M scale-up	<code>run_mllm.py</code>	~85 min

4.10 Discussion

The ablation paints a sharper picture than the headline number alone. Gated fusion accounts for most of the empirical RMSE change; the ordinal head provides native probability estimates cheaply but does not help RMSE on its own. This interaction suggests that ordinal heads are most useful here when paired with side-information mechanisms, not as a stand-alone substitute for them.

Limitations. The benchmark is small (100K ratings); the side information is limited to demographics and genres; and the ordinal head is calibrated to the rating-class marginal but does not model per-user heteroscedasticity.

5 Conclusion

We presented a calibration recipe for cumulative-link collaborative filtering: initialise the monotone thresholds from the training-set rating quantiles, parameterise their spacing through a softplus-inverse, and exclude them from L_2 regularisation. Paired with per-dimension gated side information, the resulting model attains RMSE 0.9124 ± 0.0036 and NLL 1.2584 across the five MovieLens-100K splits, improving the across-split mean over both a no-fusion MF analogue and additive fusion. The main lesson is that gating drives the RMSE gain, while the ordinal head gives calibrated probabilities and better boundary-class behaviour at negligible parameter cost. MovieLens-1M repeats the same ordering at larger scale, and the cold-start breakdown shows that the gated advantage is strongest in the mid-tail rather than uniform across users.

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